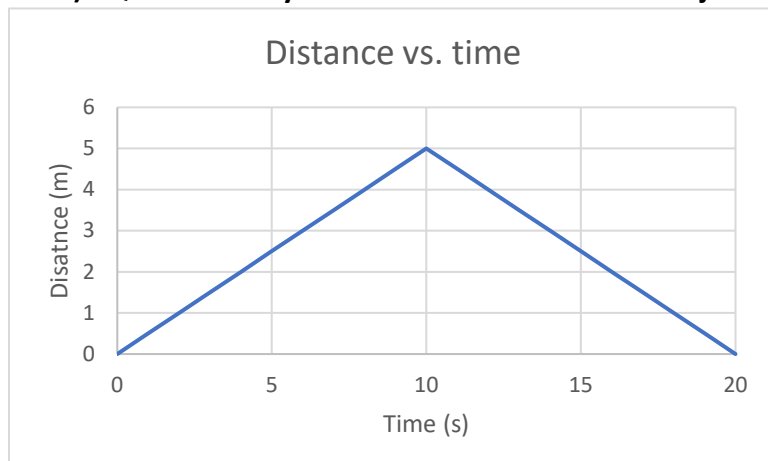


Graphs of Constant Speed Motion: Matching Graphs to Motion Scenario

Synopsis: Motion graphs *quantitatively* encapsulate the motion of an object – we can learn the position and speed of the object as a function of time. In this Note, you are tasked to 1) Describe the motion in words from a motion graph (distance vs. time or speed vs. time, 2) Sketch a motion graph from a description of an object's motion.

Performance Expectations: Give a quantitative description of an object's motion from a motion graph. Draw the motion graph from a description of an object's motion.

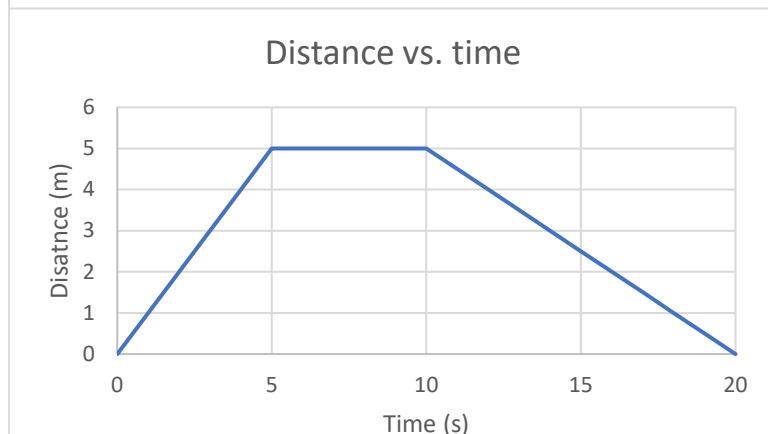
1) Quantitatively describe the motions of an object in the graphs (Must answer questions)



How far? Forward or backward? How fast?

- 0 – 10 s:

- 10 – 20 s:

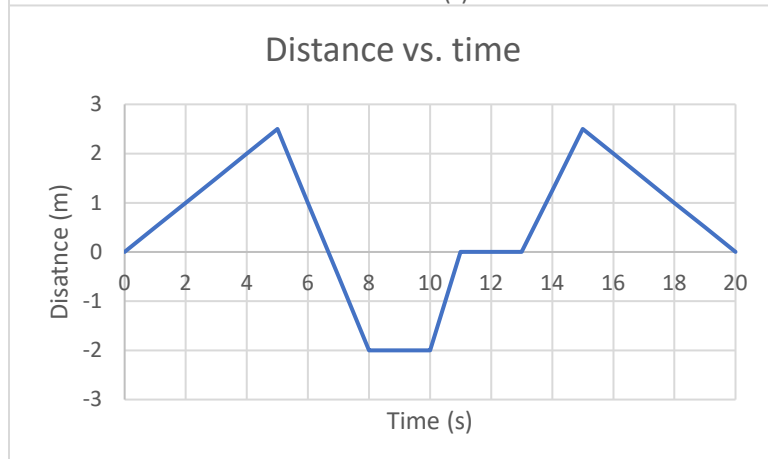


How far? Forward or backward? How fast?

- 0 – 5 s:

- 5 – 10 s:

- 10 – 20 s:



How far? Forward or backward? How fast?

- 0 – 5 s:

- 5 – 8 s:

- 8 – 10 s:

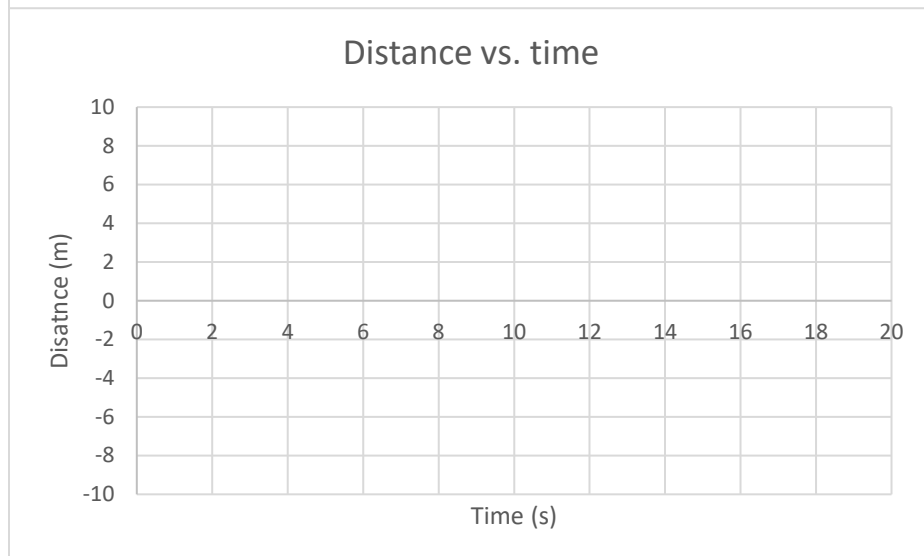
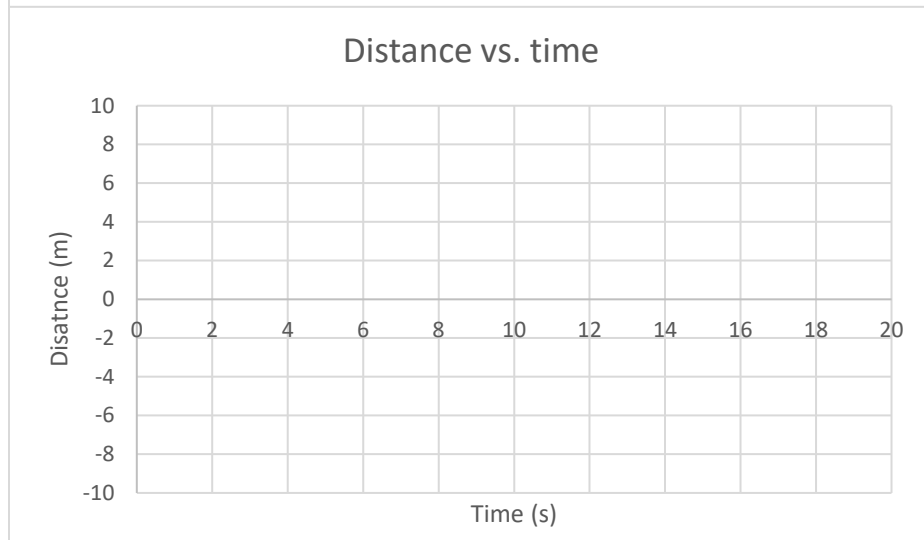
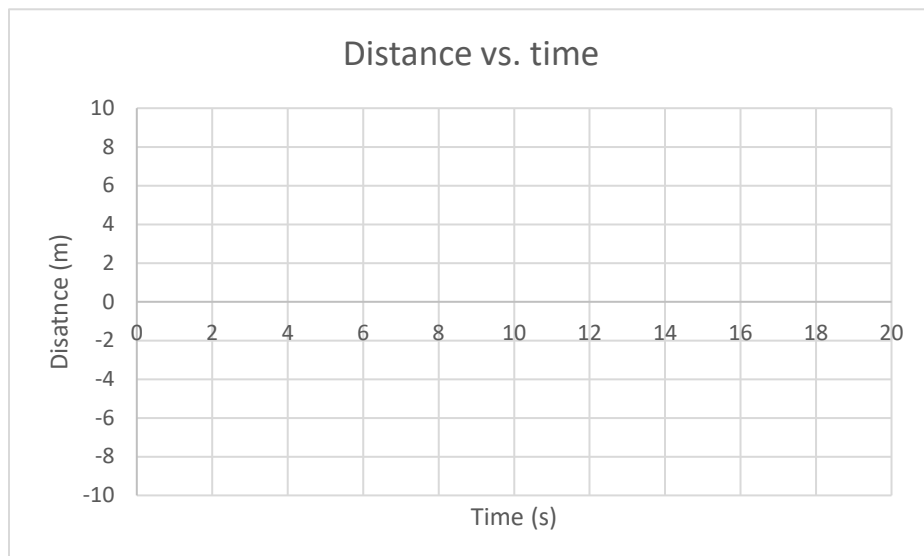
- 10 – 11 s:

- 11 – 13 s:

- 13 – 15 s:

- 15 – 20 s:

2) Draw the graph matching the description of the motion.



- Move forward at 1 m/s for 2s.
- Stop for 2s.
- Move forward at 2 m/s for 2s.
- Stop for 4 s.
- Move forward at 0.5 m/s 4s.
- Move backward at 1 m/s for 6 s.

Q) Where am I?

- Move backward at 1 m/s for 4s.
- Stop for 4 s.
- Move forward at 2 m/s for 4s.
- Stop for 2 s.
- Move backward 4 m in 6 s.

Q) Where am I?

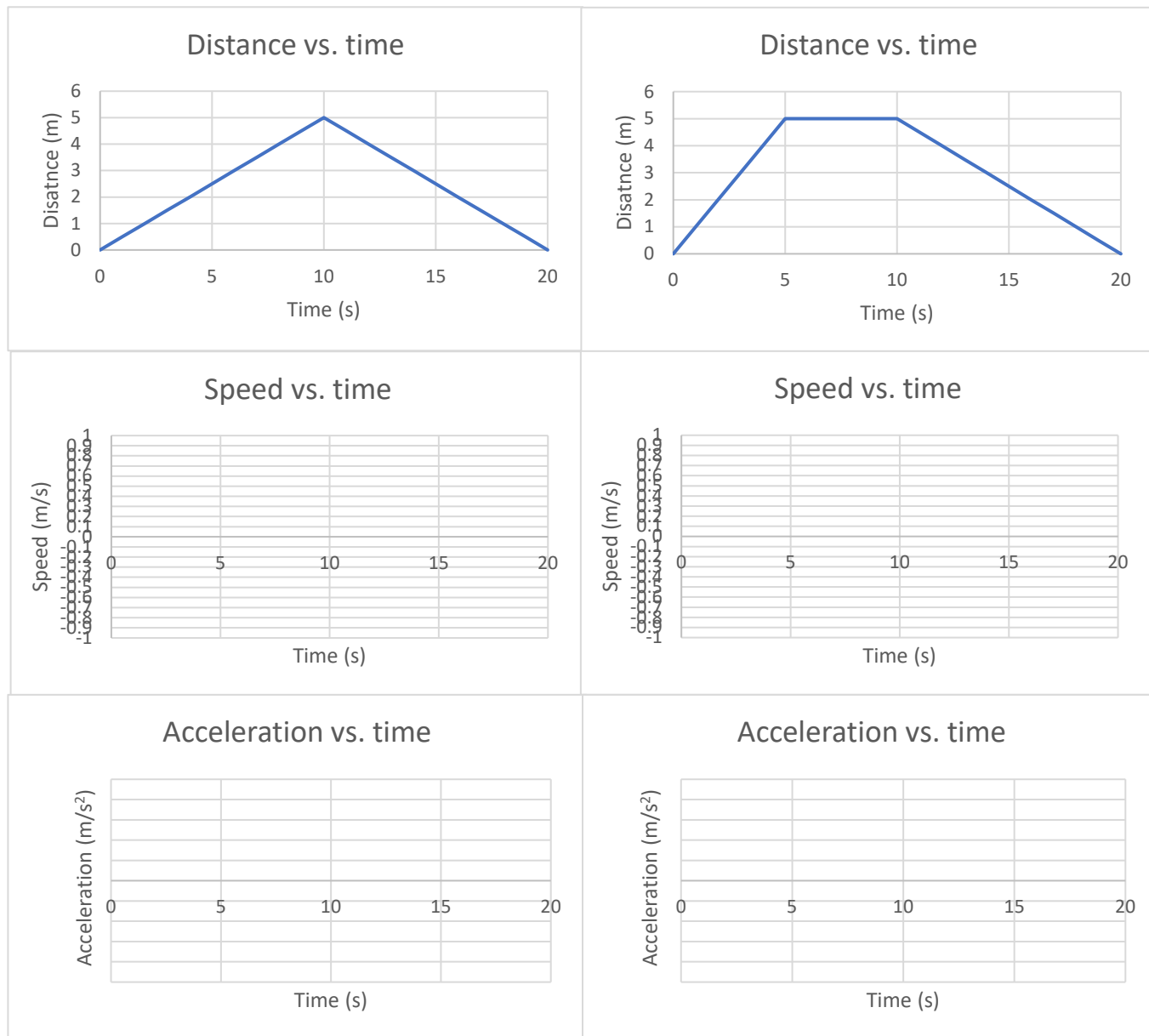
Q) What is my final speed?

- Move forward at 2 m/s for 4s.
- Stop for 2 s.
- Move backward at 2 m/s for 8 s.
- Move forward at 1 m/s for 6 s.

Q) Where am I?

Q) What is my final speed?

3) Draw corresponding speed vs. time and acceleration vs. time graphs.



In Calculus Notation

The instantaneous velocity at any moment is the derivative of position, $v(t) = dx/dt$ — the slope of the tangent line to the distance-vs-time graph at that instant. Position is recovered by integrating velocity:

$$x(t) = x_0 + \int_0^t v(t') dt'$$

Geometrically, that integral is the running signed area under the velocity-vs-time graph you sketched above. Show this for one of your scenarios: